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Comparison of shoulder kinematics between Theia 3D markerless motion capture and marker-based motion capture during full arm ranges of motion

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ABSTRACT

While markerless motion capture system has recently gained interest, little is known on the accuracy of Theia 3D for measuring shoulder kinematics. This study aimed to compare shoulder kinematics across the full range of arm motion between Theia 3D and a gold standard marker-based motion capture system. Shoulder kinematics of 20 healthy subjects were measured during three repetitions of bilateral full arm scaption, abduction, flexion, internal rotation, external rotation, and internal and external rotation at 90° abduction. Data were simultaneously collected with Theia 3D (v2024.1.24) markerless and marker-based motion capture. Markerless and marker-based angular trajectories had similar patterns, with larger differences towards the inflection point of the movements and with the markerless trajectories having mostly greater values than marker-based trajectories. Root mean square differences were smallest for abduction angles (<6°) and largest for the external rotation angles (15.3°–22.3°). Coefficients of multiple correlations (CMCs) were mostly good to excellent (>0.75), but CMC was moderate (0.65–0.75) for flexion angles during rotation at 90° and weak (<0.65) for flexion angles during external rotation. For almost all tasks, the ranges of motion differed significantly between the two measurement systems in all three planes. However, mean differences in the coronal plane and in the sagittal plane were within the minimal clinically important differences. We found moderate to very strong correlations between the range of motion parameters of the two measurement methods. The assessment of shoulder kinematics with Theia 3D is promising but further improvements are needed for clinical routine application.

1. Introduction

The rotator cuff muscles are the primary dynamic stabilisers of the glenohumeral joint, allowing a large range of motion (ROM) (Veeger and van der Helm, 2007) that is essential for many daily activities. However, shoulder musculoskeletal disorders have an incidence rate of 17.9/1000 per year and may limit shoulder functionality (Iavernig et al., 2023). Thus, an accurate assessment of the shoulder ROM is important for the clinical evaluation, training or rehabilitation.

In clinical practice, shoulder function is usually assessed as a combination of range of arm motion in all planes and patient-reported outcome measures, most commonly represented by the Constant score (Constant et al., 2008; Constant and Murley, 1987), the American Shoulder and Elbow Surgeons Shoulder score (Richards et al., 1994), or the Disabilities of Arm, Shoulder and Hand questionnaire (Hudak et al., 1996). However, these scores have a subjective component and usually do not provide objective data on specific movement characteristics (e.g. speed, deceleration time, or ROM) (De Baets et al., 2017; Longo et al., 2022). Shoulder kinematic analysis can help to obtain such objective data and provide relevant information about functional impairment and be used to monitor rehabilitation progress.

The current gold standard for non-invasive and radiation-free measurements of shoulder kinematics is marker-based motion capture (Hansen et al., 2024), but its applicability in clinical settings is limited due to long preparation and post-processing times, and because marker placement may depend on the examiner (Chiari et al., 2005; Lahkar

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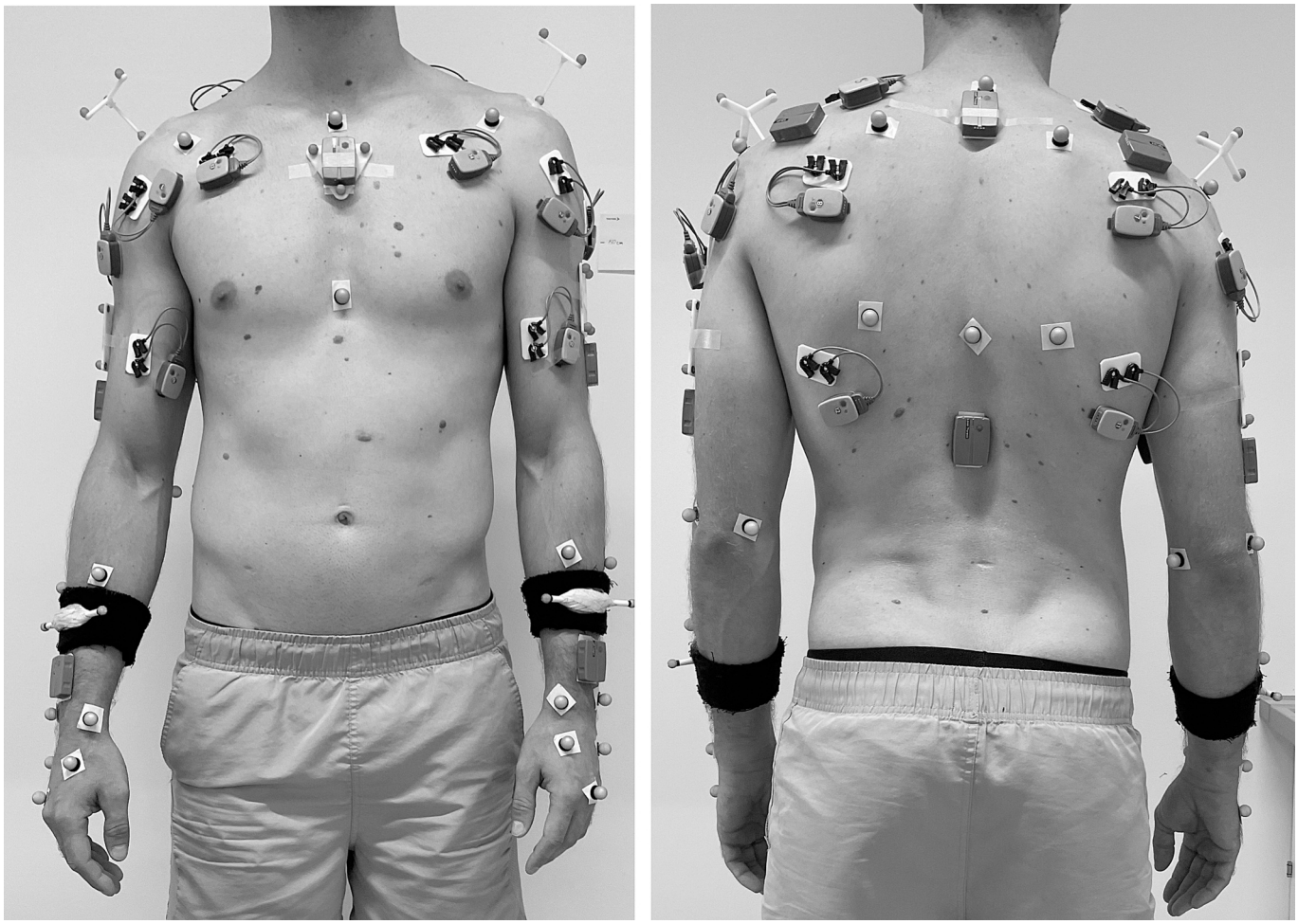


Fig. 1. Marker placement, anterior (left) and posterior (right) view. Electromyography and inertial measurements unit sensors were also attached but not analysed in this study.

et al., 2022). To overcome these limitations, markerless motion capture systems, like Theia 3D, have recently gained interest. Theia 3D estimates the 3D position of body segments from videos using deep convolutional neural networks and inverse kinematics. Because minimal preparation is required, data acquisition with a markerless motion capture system requires little time and is thus more feasible in a clinical setting. Although the scapula is not modelled in Theia 3D, it may remain valuable for the clinical assessment of patients with shoulder or neurological conditions and in sports to monitor training and progress by providing objective humerothoracic kinematics. The accuracy of Theia 3D compared to marker-based motion capture for measuring kinematics during gait has been investigated extensively with good results (D'Souza et al., 2024; Kanko et al., 2021b, 2021a; Outerleys et al., 2024; Wren et al., 2023). In contrast, such accuracy for assessing shoulder kinematics has only been investigated in adults during boxing (Lahkar et al., 2022) with root mean square differences (RMSDs) for shoulder angles of 6.3° to 12°; in adults during running with a change of direction (Edwards et al., 2024) with RMSDs for shoulder angles of 6.7° to 56.6°; and in children during a box and block test (Hansen et al., 2024) with RMSDs for all upper extremities joint angles below 6°. As these studies analysed specific movements with a limited ROM, further research is needed to determine how Theia 3D performs during the full range of arm motion in all planes, including overhead movements. Thus, this study aimed to compare shoulder kinematics across the full range of arm motion as measured by the Theia 3D markerless motion capture system and by a gold-standard marker-based motion capture system in healthy adults.

2. Methods

2.1. Participants

A total of 20 young control subjects participated in this prospective comparative study. Subjects were included if they were between 18 and 30 years of age and had no previous known elbow or shoulder injuries or symptoms. Exclusion criteria were: a body mass index (BMI) greater than 35 kg/m^2, inability to provide informed consent, known pregnancy, implanted pacemakers, and the presence of any neuromuscular disorders or other pathologies affecting upper limb movement. The criteria were checked by anamnesis. Written informed consent was obtained from all subjects prior to data collection. This study was approved by the regional ethics board (Ethics Committee Northwestern and Central Switzerland, EKNZ 2024–00479) and conducted in accordance with the Declaration of Helsinki.

2.2. Study procedure and data collection

All subjects were equipped with anatomical reflective markers on landmarks of the upper extremities and torso according to the International Society of Biomechanics (ISB) guidelines (Wu et al., 2005). Additional cluster markers were placed on the upper arms, acromion and torso (Fig. 1).

Synchronised markerless (sampling rate 100 Hz, Miqus cameras, Qualisys AB, Gothenburg, Sweden) and marker-based (sampling rate 200 Hz, Arqus cameras, Qualisys AB, Gothenburg, Sweden) motion

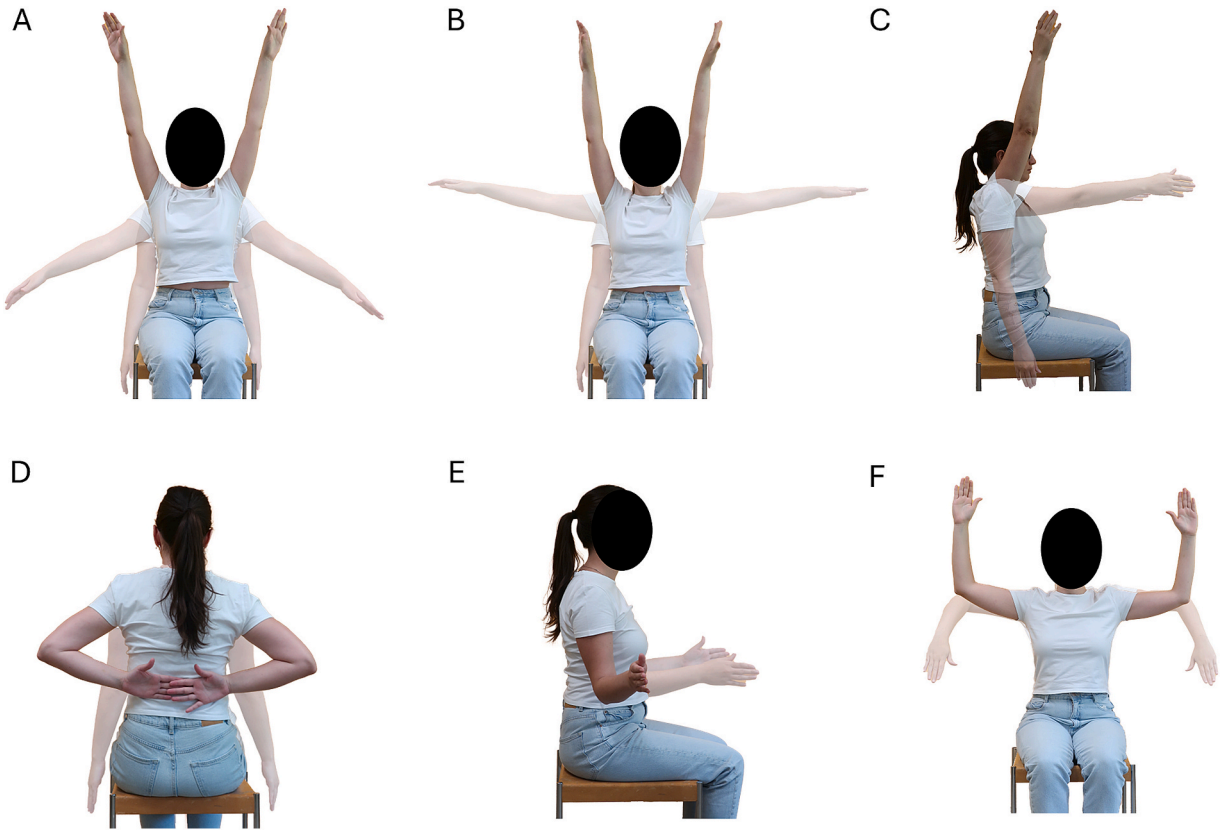


Fig. 2. Shoulder tasks. Scaption (A), abduction (B), flexion (C), internal rotation (D), external rotation (E), and internal and external rotation at 90° abduction (F).

capture data were collected during three repetitions of bilateral full arm scaption, abduction, flexion, internal rotation, external rotation, and internal and external rotation at 90° abduction (Fig. 2). Full arm scaption, abduction, and flexion were performed with elbows extended and hands in neutral position from the starting position with the arms hanging at the side to the maximum ROM and back. For the internal rotation, subjects were asked to reach from the starting position, with the arms hanging at the side, to the highest reachable point along the midline of the back and back. For the external rotation, subjects were asked to flex their elbow to 90° from the starting position with their arms hanging at the side and then rotate their arm externally to their maximum ROM and internally back to the starting rotation. Lastly, subjects were asked to rotate their arm in 90° abduction with 90° elbow flexion first internally and then externally to its maximum ROM and then internally back to the neutral rotation. All tasks were performed with the subjects seated on a stool. Participants were given both verbal and visual instructions on how to perform the movement tasks, and correct task execution was qualitatively assessed before and during the recording of the trials. An additional static trial was recorded with the subject seated upright and with the arm hanging at the side with the thumbs pointing anteriorly.

2.3. Data processing

Markerless data were processed using Theia 3D (v2024.1.24, Theia Markerless Inc., Kingston, ON, Canada). Shoulder joints were allowed 6 degrees of freedom and 3D poses were filtered using a spline low-pass Butterworth filter with a cut-off frequency of 6 Hz (Song et al., 2023). The output data from Theia 3D were then further processed in Matlab (Version 2024b, MathWorks Inc., Natick, MA, USA). Rotation matrices from Theia 3D (X-axis pointing laterally, Y-axis anteriorly, Z-axis cranially) were adjusted to match the ISB coordinate systems (X-axis pointing anteriorly, Y-axis cranially, Z-axis laterally) (Wu et al., 2005).

For the marker-based motion capture, marker trajectories were reconstructed in Qualisys Track manager (v2023.3, Qualisys AB, Gothenburg, Sweden) and further processed in Matlab. Marker data were down-sampled to match the sampling rate of the markerless data (100 Hz) and filtered with a low-pass zero-lag fourth-order Butterworth filter with a cut-off frequency of 6 Hz (Winter, 2009). Glenohumeral joint centres were estimated in the static trial using the regression equation outlined by the International Shoulder Group (Campbell et al., 2009), which is based on five scapular landmarks (trigonium spinae, angulus inferior, angulus acromialis, processus coracoideus, acromioclavicular joint). Least squares algorithms were used to reduce skin movement artefacts (Challis, 1995).

For both systems, humerothoracic joint kinematics were reconstructed for all tasks using the X-Z-Y sequence (corresponding to abduction, flexion, and external rotation), except for full flexion, for which the Z-X-Y sequence was used (corresponding to flexion, abduction, and external rotation) (Lempereur et al., 2014). Unphysiological spikes in markerless kinematics observed in some cases were identified in Matlab as outliers of the differences between adjacent frames, removed and gaps were filled with a spline function (modified Akima piecewise cubic Hermite interpolation, 'makima' in Matlab).

The events of the three repetitions of each task (start, maximum exertion, end) were then identified using the minimum and maximum humerothoracic angles from the marker data in the primary plane of motion of the task (abduction angle for scaption and abduction tasks; flexion angle for flexion task; and external rotation angle for all rotation tasks). These events identified using the marker data were also applied to the markerless data. Each repetition was then time-normalised with 101 data points from the starting position to maximum exertion and with 101 data points from maximum exertion to the end position. The average waveforms for the humerothoracic angles in each plane were calculated from the three repetitions for each shoulder for all tasks.

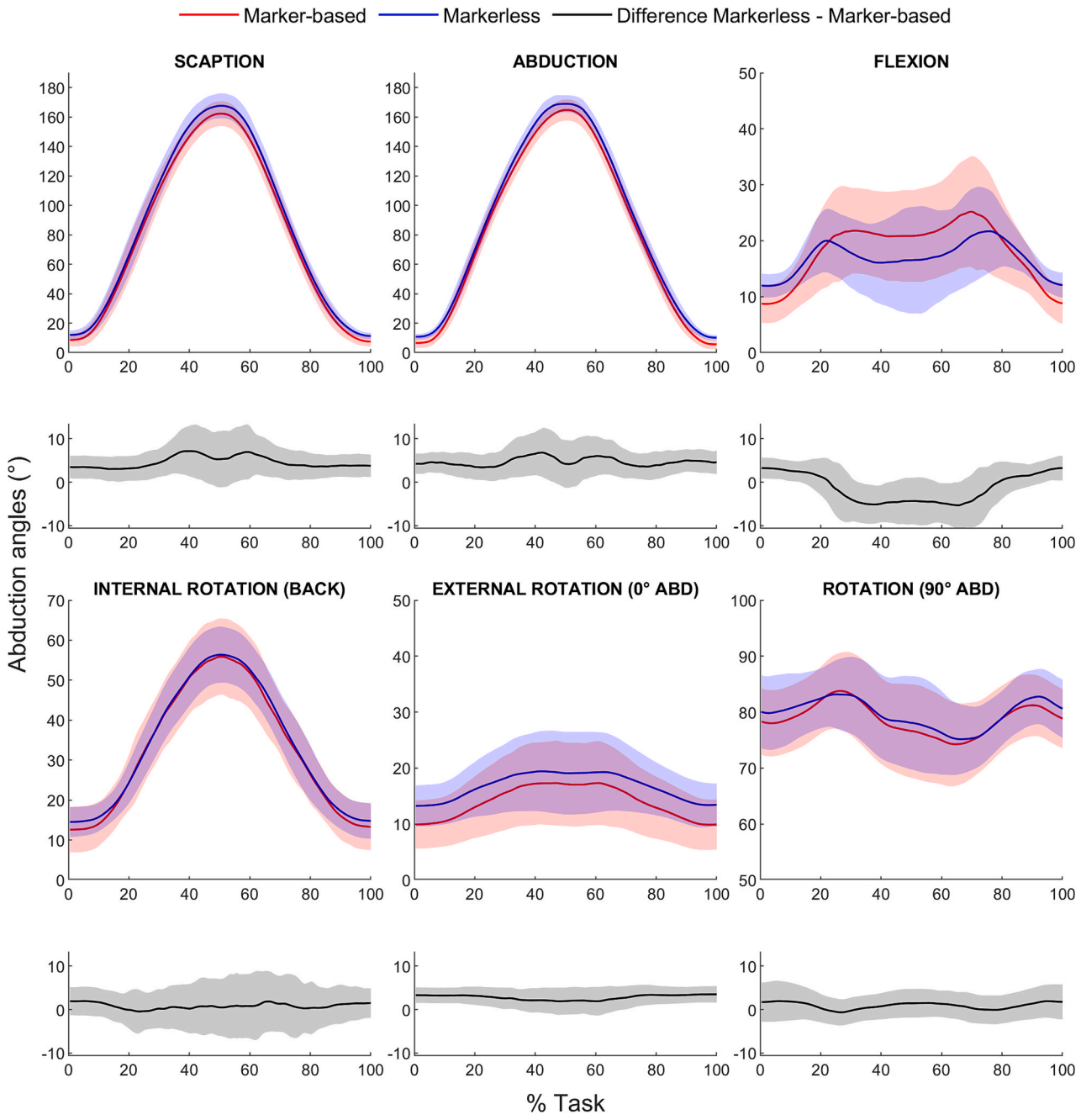


Fig. 3. Mean (standard deviation) marker-based and markerless abduction angle waveforms and their difference for all tasks analysed. ABD, abduction.

2.4. Statistical analysis

All statistical analyses were performed in Matlab. Humerothoracic joint kinematics calculated from the markerless system were compared with the reference marker-based system using RMSD and coefficient of multiple correlation (CMC) (Ferrari et al., 2010). All analyses were performed for each angle and each movement task based on the average waveforms of all shoulders (n = 40). The CMC measures the similarity of the waveforms by taking into account the offset differences of the two simultaneous measurements and was interpreted as weak (CMC < 0.65), moderate (0.65 ≤ CMC < 0.75), good (0.75 ≤ CMC < 0.85), very good (0.85 ≤ CMC < 0.95), and excellent (CMC ≥ 0.95) (Ferrari et al., 2010). In addition, paired t-tests were used to detect differences in the measurements of ROM (i.e. absolute difference between peaks values) in all planes for each of the tasks between the markerless and the marker-based systems. Limits of agreements (Bland and Altman, 1986) of these discrete measurements were also analysed. Pearson's correlation coefficients were calculated for the ROM in all planes measured with both measurement methods for each task. Correlations were considered very weak (R < 0.2), weak (0.2 ≤ R < 0.4), moderate for (0.4 ≤ R < 0.6), strong (0.6 ≤ R < 0.8), and very strong (R ≥ 0.8) (Salkind, 2017). The significance level was set at $P = 0.05$. Minimal clinically important differences (MCIDs) of 7° for abduction, 12° for flexion, and 3° for external rotation (Simovitch et al., 2018) were used to determine the clinical relevance of the angular differences between the two measurement methods.

It was estimated that a minimum of 17 subjects would be required to detect a strong correlation of R = 0.7 with 90 % power and a two-tailed significance level of P < 0.05.

Table 1
Assessment parameters (mean ± standard deviation) of the humerothoracic joint angles in all planes for all tasks.

Task	Abduction angles		Flexion angles		External rotation angles	
	RMSD (°)	*CMC*	*RMSD (°)*	*CMC*	*RMSD (°)*	*CMC*
Scaption	6.0 ± 1.8	0.997	8.1 ± 1.9	0.836	21.0 ± 4.5	0.919
Abduction	5.9 ± 1.2	0.997	8.2 ± 3.2	0.822	22.3 ± 4.7	0.897
Flexion	5.4 ± 1.5	0.770	9.1 ± 2.7	0.995	15.6 ± 5.6	0.871
Internal rotation	5.3 ± 1.4	0.971	11.3 ± 0.8	0.788	19.2 ± 6.1	0.793
External rotation	3.7 ± 0.2	0.755	9.3 ± 0.9	0.439	17.0 ± 8.5	0.879
External rotation at 90° abduction	3.6 ± 0.6	0.856	9.1 ± 0.4	0.685	15.3 ± 5.7	0.967

RMSD, root mean square difference; CMC, correlation of multiple correlation

3. Results

This study sample included 10 men and 10 women (mean ± standard deviation age: 27.1 ± 1.9 years; height: 173.3 ± 10.7 cm; body mass: 70.3 ± 12.2 kg; BMI: 23.2 ± 2.0 kg/m^2. All participants completed all tasks with both arms, and thus 40 shoulders were included in the analyses. Each repetition lasted 5.2 ± 1.0 s for scaption, 5.1 ± 1.0 s for abduction, 5.1 ± 0.8 s for flexion, 4.2 ± 0.7 s for internal rotation, 3.1 ± 0.5 s for external rotation, and 3.8 ± 0.5 s for external rotation at 90° of abduction.

3.1. Coronal plane

The markerless and marker-based abduction angles waveforms had qualitatively similar patterns throughout the entire movements for all tasks where the markerless waveforms were shifted towards higher abduction angles compared to marker-based waveforms (Fig. 3). The greatest differences between the two systems were observed during the flexion task, where the markerless waveform did not consistently have higher values than the marker-based waveforms but had lower values between approximately 20 % and 80 % of the tasks. This is also reflected in the moderate CMCs and reliability (Table 1), but the RMSD of the abduction angle in the flexion task was on average 5.4°. Of the tasks analysed, the external rotation and rotation at 90° abduction tasks had the lowest RMSDs between the two waveforms with values below 4° and good to excellent CMCs (Table 1). Slightly higher RMSDs were found for the other tasks, but still less than 6° (within the MCID) and with excellent CMCs (Table 1).

Significant differences between markerless and marker-based ROMs with respect to the coronal plane (abduction/adduction angles) were found only for the flexion and external rotation tasks ($P \leq 0.001$). However, the mean differences were smaller than 5° (within the MCID), with the markerless angles being smaller than the marker-based ones (Table 2). The limits of agreements between systems for the ROM were smallest for the external rotation task at 90° abduction (within 6.2°) and largest for the flexion task (within 20.3°), with only a weak correlation measured for this task. All other tasks had strong to very strong correlations between the markerless and marker-based ROMs (Table 2).

3.2. Sagittal plane

Waveforms of the flexion angles for the tasks analysed were quite similar between the markerless and marker-based systems, with the markerless systems tending to have higher values than the marker-based systems. In contrast, for the scaption and adduction tasks, the opposite occurred from approximately 30 to 70 % of the tasks (Fig. 4). Overall, the average RMSDs between the two systems were between 8.1° and 11.3° (below the MCID) for the tasks analysed (Table 1). However, the

Table 2
Assessment parameters for the comparison between marker-based and markerless range of motion of the humerothoracic joint in all planes for all tasks.

	Scaption	Abduction	Flexion	Internal rotation	External rotation	External rotation at 90° abduction
Coronal plane						
Marker-based ROM (°), mean ± SD	155.1 ± 8.9	159.6 ± 8.3	20.9 ± 7.8	44.2 ± 10.5	9.6 ± 5.2	14.9 ± 4.7
Markerless ROM (°), mean ± SD	156.8 ± 8.5	159.7 ± 6.5	16.0 ± 6.8	42.8 ± 7.5	8.1 ± 5.3	14.3 ± 5.5
Mean difference (°)[LoA]	1.8[−11.3; 14.9]	0.2[−11.9; 12.3]	−4.9[−20.3; 10.5]	−1.3[−13.1; 10.4]	−1.4[−6.5; 3.6]	−0.6[−6.2; 4.9]
P-value[a]	0.101	0.863	<0.001	0.168	0.001*	0.156
Pearson's correlation[b]	0.703	0.674	0.428	0.827	0.880	0.857
Sagittal plane						
Marker-based ROM (°), mean ± SD	32.4 ± 11.3	33.5 ± 12.9	181.3 ± 12.4	30.8 ± 10.8	16.8 ± 7.1	23.9 ± 6.1
Markerless ROM (°), mean ± SD	24.6 ± 6.4	22.4 ± 7.2	173.1 ± 12.3	31.9 ± 9.7	14.2 ± 6.5	23.1 ± 4.8
Mean difference (°)[LoA]	−7.8 [−22.4; 6.9]	−11.1[−29.0; 6.9]	−8.2 [−23.4; 7.0]	1.0[−6.5; 8.6]	−2.6[−8.8; 3.7]	−0.8[−8.4; 6.9]
P-value[a]	<0.001	<0.001	<0.001	0.232*	<0.001	0.225
Pearson's correlation[b]	0.783	0.721	0.802	0.935	0.892	0.769
Transverse plane						
Marker-based ROM (°), mean ± SD	100.6 ± 13.6	97.7 ± 14.3	66.0 ± 14.6	39.5 ± 13.4	59.1 ± 9.4	106.7 ± 11.4
Markerless ROM (°), mean ± SD	109.7 ± 12.7	108.2 ± 16.3	58.2 ± 12.4	73.0 ± 24.6	85.4 ± 11.0	149.9 ± 12.5
Mean difference (°)[LoA]	9.1[−12.0; 30.2]	10.5[−16.8; 37.8]	−7.8[−29.0; 13.4]	33.5[1.9; 65.1]	26.3[10.9; 41.7]	43.2[29.7; 56.7]
P-value[a]	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Pearson's correlation[b]	0.665	0.593	0.690	0.794	0.713	0.838

SD, standard deviation; ROM, range of motion; LoA, limits of agreement.
[a] P-values of paired *t*-test.
[b] all P-values ≤ 0.006 of Pearson's correlation
* P-value of Wilcoxon sign rank test.

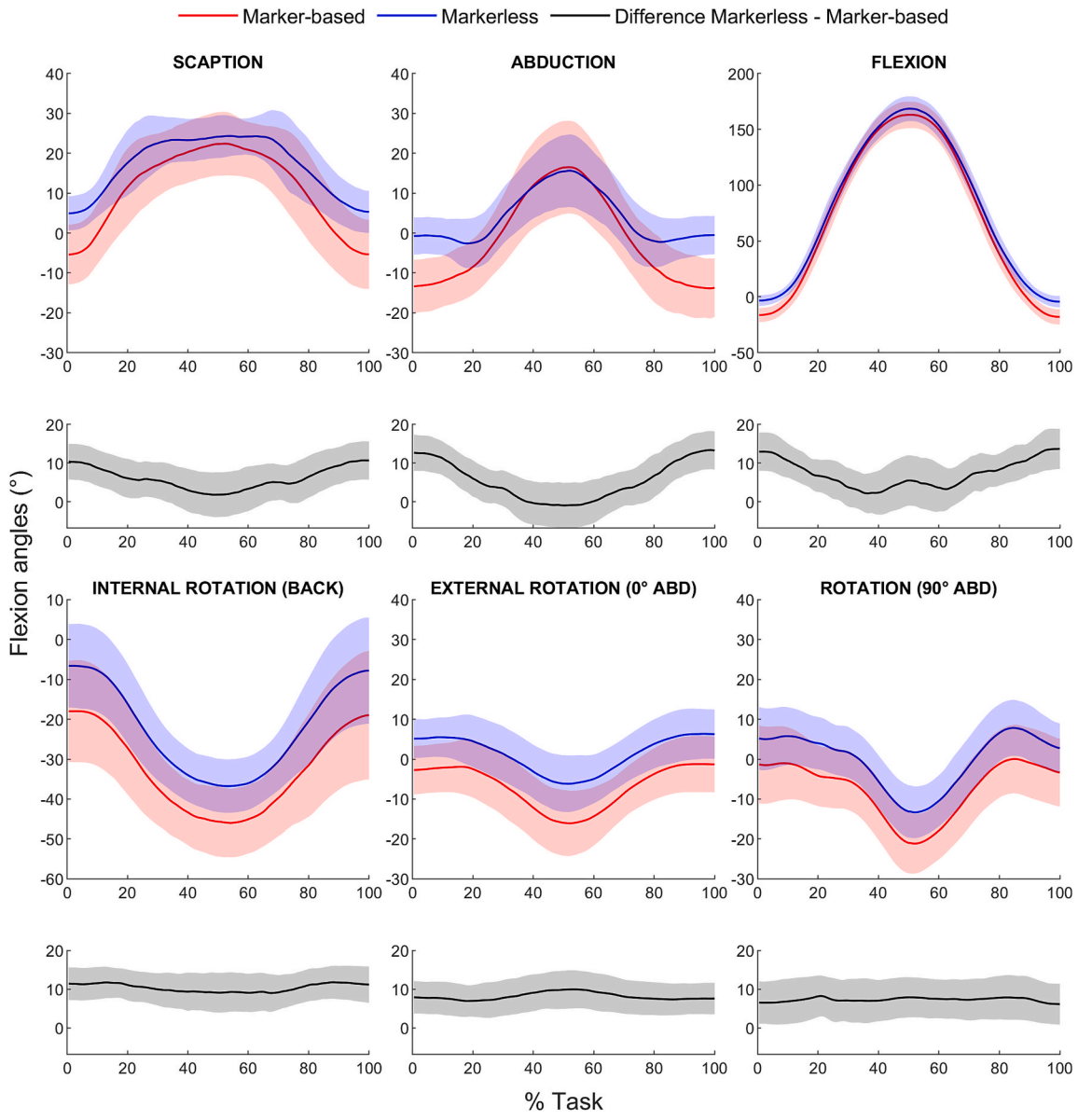


Fig. 4. Mean (standard deviation) marker-based and markerless flexion angle waveforms and their difference for all tasks analysed. ABD, abduction.

CMC was excellent for the flexion task, moderate to good for the scaption, abduction and external rotation at 90° arm abduction tasks, but weak for the external rotation task (Table 1).

The markerless ROMs with respect to the sagittal plane (flexion/extension angles) were statistically significantly different from the marker-based ROMs for all tasks ($P < 0.001$), except for internal rotation ($P = 0.232$) and external rotation at 90° abduction ($P = 0.225$, Table 2). The largest difference in ROM measurement was observed in the abduction task, with the markerless system showing an average of 11.1° less ROM than the marker-based system (Table 2). However, these differences were still within the MCID for the flexion angles. The limits of agreement were greater for the scaption, abduction and flexion tasks (up to 29.0°) than for all rotation tasks (up to 8.8°, Table 2). For all tasks, the correlations between the two measurement systems for ROM were strong to very strong (Table 2).

3.3. Transverse plane

The markerless waveforms of the external rotation angles had similar patterns to the marker-based waveforms, with larger differences near the inflection points for all tasks analysed (Fig. 5). We found large average RMSDs, well above the MCID for external rotation angles, ranging from 15.3° for the external rotation task at 90° arm abduction to 22.3° for the abduction task. However, the CMCs were good to excellent for all tasks analysed (Table 1).

Markerless ROMs in the transverse plane differed significantly from the marker-based ROMs for all tasks ($P \leq 0.001$), with mean differences larger than the MCID and with values of up to 43.2° for the external rotation at 90° abduction task (Table 2). Compared to the marker-based ROMs, these were overestimated by the markerless system for all tasks, with the exception of the flexion task where the opposite was observed (Table 2). Large limits of agreement up to 65.1° were found for all tasks. For all tasks performed, moderate to very strong correlations were measured between the markerless and marker-based measures of ROM in relation to external/internal rotation angle (Table 2).

4. Discussion

This study compared humerothoracic kinematics simultaneously acquired using Theia 3D and marker-based motion capture during full

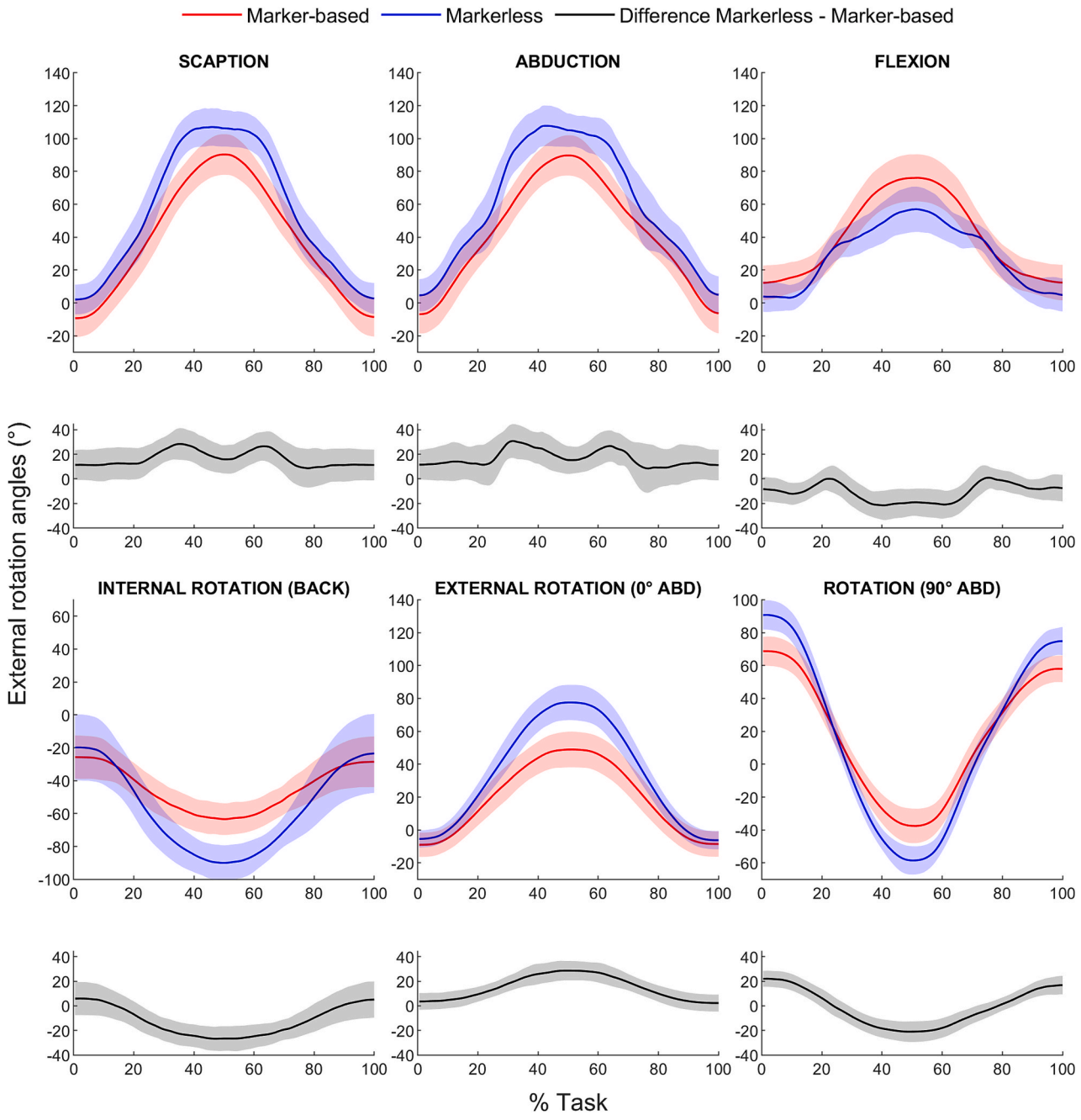


Fig. 5. Mean (standard deviation) marker-based and markerless external rotation angle waveforms and their difference for all tasks analysed. ABD, abduction.

arm range-of-motion tasks in all planes. This allowed us to gain a detailed insight into the accuracy of the Theia 3D for upper limb assessment.

Overall, we found similar shoulder kinematic patterns between the two measurement methods in all planes across all tasks with larger deviations towards the inflection points. The best agreements were found for the measurements in the coronal plane and worst for those in the transverse plane. Abduction angles had indeed the smallest RMSDs and the highest CMCs, while external rotation angles had the largest RMSDs, but the CMCs were higher than those for the flexion angles. These higher CMCs may be due to the tendency for larger ROMs in the transverse plane than in the sagittal plane.

Compared to other studies comparing Theia 3D with marker-based shoulder kinematics, we found larger RMSDs ($> 6°$) than Hansen et al. (Hansen et al., 2024). However, we found similar to slightly higher RMSDs for the abduction and flexion angles compared to Lahkar et al. and Edwards et al., and much smaller RMSDs for the external rotation angles than Edwards et al. but approximately twice as high as Lahkar et al. (our study vs. Lahkar et al. vs. Edwards et al.; abduction: 6.6° vs. 6.6° vs. 6.7°; flexion: 11.5° vs. 10° vs 15.7°; external rotation: 22.4° vs. 12° vs. 56.6°, respectively) (Edwards et al., 2024; Lahkar et al., 2022). A possible explanation for the larger RMSDs in our study could be the greater ROM required for our tasks (abduction: 0° to 180°; flexion: −50° to 180°; and external rotation −80° to 100°) compared to the other studies (abduction: −20° to 70°, flexion: −40° to 110°, and external rotation angles: −0° to 50°) (Edwards et al., 2024; Hansen et al., 2024; Lahkar et al., 2022).

We observed larger discrepancies between the two measurement methods at the extremes of the tasks. Consequently, almost all ROM measurements were statistically different between Theia 3D and marker-

based motion capture in the sagittal and transverse planes. It is therefore possible that extreme positions are under-represented in the Theia 3D training dataset, as to date most publications focus on lower extremity analysis that involve limited upper extremity motion (D'Souza et al., 2024; Kanko et al., 2021b, 2021a; Outerleys et al., 2024; Wren et al., 2023). The largest RMSDs of the waveforms in the transverse plane may be due to the fact that rotations around the longitudinal axis of the arm produce minimal changes in the external appearance, as the segments closely resemble cylindrical shapes, making it challenging for the markerless system to accurately quantify such rotation angles. These larger discrepancies in the transverse plane may also be a consequence of the rotation sequence used, as the external rotation angles are the tertiary rotation. We chose the rotation sequence for each task according to the predominant plane of motion (Lempereur et al., 2014). However, as the shoulder joint moves considerably in all three planes, there are no rotation sequences that allow a clear clinical interpretation for all planes of movement (Šenk and Chèze, 2006). Depending on the plane and task of interest, the markerless ROM measurements were smaller or larger than the marker-based measurements. However, most correlations were moderate to strong, indicating that there are some similarities in the measurements between the two systems. Furthermore, the mean differences between the two methods of the ROM measurements in the coronal plane and in the sagittal plane for the rotation tasks were within the MCIDs (Simovitch et al., 2018).

The markerless motion capture system is still in continuous development and major updates of Theia 3D are frequently released. Compared to the previous version of Theia 3D (v2023.1.0.3160), the current version (v2024.1.24) has clearly improved for lower extremity kinematics (Theia, 2024), but only marginally for shoulder kinematics (Supplementary Fig. S1). This underlines the need to have training dataset that are suitable for the respective application and task. Although some improvements in Theia 3D are still desirable, the results of this study are promising for the applicability of Theia 3D not only in clinical evaluation, but also in monitoring rehabilitation progress of and sports performance assessment. Currently, Theia 3D provides an accurate enough assessment of the shoulder kinematics in the coronal plane but is increasingly less accurate in the sagittal and transverse planes. Hence, Theia 3D is mostly suitable when the focus of clinical evaluation, training or rehabilitation is on the abduction angle.

This study has several strengths and limitations. The strength of this study is the well-defined study protocol for comparing shoulder kinematics across the full ROM with synchronised and simultaneous markerless and marker-based data acquisition. Shoulder kinematics measured with the marker-based system were used as reference. Although marker placement was always checked by the same person with adequate palpation skills, some variability cannot be excluded, and these data may not represent true shoulder kinematics due to soft tissue artefacts and be different from the ground truth. Hence, we cannot exclude the possibility that the level of agreement of Theia 3D with the true shoulder kinematics could be slightly higher than with the marker-based system. However, as the markerless system is sensitive to the quality of the video data (e.g. lighting, contrast, resolution), the results may vary if recorded in a less controlled environment or with different clothing. As with marker-based data, markerless data also require quality checks and cleaning, hence results should be carefully interpreted. Results may also be different in subjects with a higher BMI or with conditions that affect the kinematics of the upper extremity. Only subjects without known shoulder or elbow disorders were included in this study, but we cannot exclude the possibility of rotator cuff tears or tendinopathies in some shoulders although its likelihood should be rather low.

5. Conclusion

Theia 3D markerless motion capture produced overall kinematic patterns for the shoulder joint that were similar to those measured with marker-based motion capture. This highlights the potential for clinical applications of Theia 3D, particularly in situations where the identification of movement patterns is critical. However, the level of agreement between the two systems varies depending on the movement performed, with a tendency for greater differences at the extremes of the ROM and for measurements in the transverse plane. The assessment of shoulder kinematics with Theia 3D is promising but some improvements are still needed to increase the agreement with the gold standard marker-based motion capture system. At present, Theia 3D is sufficiently accurate compared to gold-standard marker-based motion capture only for shoulder kinematics in the coronal plane but not yet in the other planes.

IRB/Ethical committee approval

Ethics Committee Northwestern and Central Switzerland EKNZ 2024–00479.

CRediT authorship contribution statement

Eleonora Croci: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Christoph Künzel:** Writing – review & editing, Investigation, Formal analysis. **Dominic Gehring:** Writing – review & editing, Supervision. **Andreas Marc Müller:** Writing – review & editing, Supervision. **Annegret Mündermann:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbiomech.2025.112905.

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